Investigating the Ground State Energies of Three-Electron Systems

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The variational method with a Hylleraas-style wavefunction is applied to estimate the ground state energy of helium, before being extended to lithium. A ground state energy of lithium of -7.470 E_h is obtained through this method, which is closer to the the accepted value of -7.478 E_h than the value obtained through the use of the Slater determinant method of -7.419 E_h, but is not the same as that obtained in the corresponding literature. We also test a method of determining the elements of the Hamiltonian and unity matrices in the context of the variational method applied to Hylleraas-style wavefunctions.

I. Introduction and Theory

In this work, we investigate the application of the variational method using Hylleraas-style wavefunctions to the helium atom: the simplest system in which we must consider the mutual interactions between electrons as well as the electron-nucleus interactions. We consider calculating the ground state energy of lithium, which contains both closed and open shells. Such systems provide a test of the effectiveness of computational methods which could then be used to predict the properties of more complicated systems.

While the ground state energy of a two-particle system such as hydrogen can be solved for analytically, this is not the case for systems with three or more particles. The simplest such system to consider would be the helium atom, for which we can use a method first described by Hylleraas in 1929.[1]

The Hamiltonian of an n-electron system is

$$\hat{H} = \sum_{i=0}^{n}\left[-\frac{1}{2}\nabla_i^2 - \frac{Z}{r_i}\right] + \sum_{i=1}^{n}\sum_{j>i}^{n}\frac{1}{r_{ij}}, \tag{1}$$

where it should be noted that this accounts for only the nucleus-electron and electron-electron interactions, neglecting any fine structure effects.

It can be shown[2] that the functional

$$E[\phi] = \frac{\langle\phi|\hat{H}|\phi\rangle}{\langle\phi|\phi\rangle}, \tag{2}$$

where ϕ is the trial wavefunction, satisfies the relationship $E_0 \leq E[\phi]$, such that the result of minimising the functional is to approximate the system's ground state energy E_0.

In order to choose the basis states for the trial wavefunction, we note that the ground state energy corresponds to a wavefunction with even spatial parity (parahelium). We then choose the basis states[3]

$$\tilde{\Phi}_{jkm}(\vec{r}_1,\vec{r}_2) = (r_1+r_2)^j(r_1-r_2)^k r_{12}^m \exp\left[-\frac{Z}{\kappa r_0}(r_1+r_2)\right], \tag{3}$$

where r_0 is the Bohr radius, κ is a variational parameter, and j, k, and m are integers, where even spatial parity requires that k is even. Each r_i represents the distance between the ith electron and the nucleus, and r_{12} is the distance between the two electrons. We can write the trial wavefunction as a linear combination of these basis states (the Hylleraas-style wavefunction):

$$\Psi(\vec{r}_1,\vec{r}_2) = \sum_{j,k,m}\tilde{a}_{jkm}\Phi(\vec{r}_1,\vec{r}_2). \tag{4}$$

We then wish to solve the Schrödinger equation

$$\boldsymbol{H}\boldsymbol{a} = E\boldsymbol{N}\boldsymbol{a}, \tag{5}$$

where $\boldsymbol{a}$ is the vector of coefficients of the terms in the wavefunction and $\boldsymbol{H}$ is the Hamiltonian matrix with elements

$$\begin{aligned} H_{ij} &= \langle\psi_i|\hat{H}|\psi_j\rangle \\ &= \langle\psi_i|\hat{T}|\psi_j\rangle + \langle\psi_i|\hat{V}|\psi_j\rangle, \end{aligned} \tag{6}$$

with ψ_n being the nth term in the trial wavefunction. $\hat{T}$ and $\hat{V}$ are the kinetic and potential energy operators respectively, and $\boldsymbol{N}$ is the unity matrix with elements

$$\boldsymbol{N} = \langle\psi_i|\psi_j\rangle. \tag{7}$$

With some algebra,[3] this is easily turned into the eigenvalue equation

$$\boldsymbol{P}\boldsymbol{z} = E\boldsymbol{z} \tag{8}$$

Where $\boldsymbol{P} = \boldsymbol{\beta}^{-\frac{1}{2}}\boldsymbol{Y}^T\boldsymbol{H}\boldsymbol{Y}\boldsymbol{\beta}^{-\frac{1}{2}}$ and $\boldsymbol{z} = \boldsymbol{\beta}^{\frac{1}{2}}\boldsymbol{Y}^T\boldsymbol{a}$. Here, $\boldsymbol{\beta}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\boldsymbol{N}$ and $\boldsymbol{Y}$ is a matrix whose columns are the eigenvectors of $\boldsymbol{N}$.

The energies of the system E can be obtained by finding the eigenvalues of $\boldsymbol{P}$ in Eq. 8, of which the minimal value corresponds to the ground state energy.

II. Application of the Variational Method to Helium

We shall first investigate the effect of increasing the number of terms in the trial wavefunction on the convergence towards the ground state energy as κ is varied. For each term, we consider systematically altering each index j, k, and m in such a way that the spatial parity requirements are not violated, i.e. that k remains even.

A logical first attempt would be to vary each index

individually up to some number n, such that (j, k, m) increases like $(0, 0, 0), (0, 0, 1), \ldots, (n, 2n, n)$. This would correspond to a trial function of $(n+1)^3$ states. The results of this choice of basis states with n increasing from 0 are indicated by the blue line in Fig. 1. (Note that, due to the chosen range of the y-axis, the result corresponding to just one basis state $(0, 0, 0)$ cannot be seen.)

It has been suggested[4] that requiring m to be odd improves the rate of convergence towards the ground state energy. Our results contradict this, as can be seen by the slower rate of convergence of the red line in Fig. 1, where we require that m is odd.

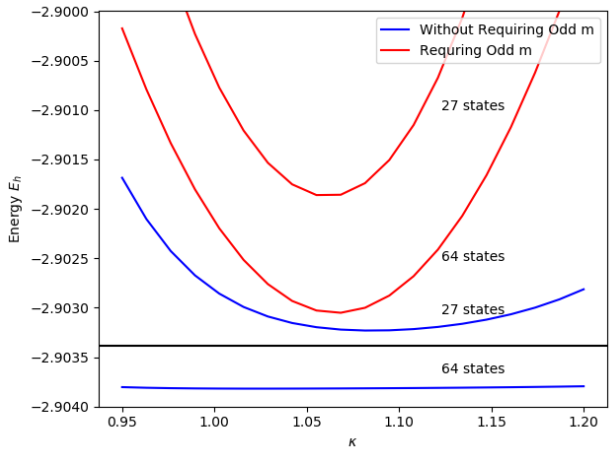

FIG. 1: The convergence of the ground state energy as we increase the number of terms in the trial wavefunction. The experimental value is shown as a horizontal black line.

As can be seen in Fig. 1, as we vary the number of terms in the trial wavefunction, we converge towards the experimentally-determined value of the ground state energy of -2.9034 E_h,[5] obtaining a value of -2.9038 E_h. We see that the convergence overshoots the experimentally-determined value, contrary to our assumption that $E_0 \leq E[\Psi]$. This is most likely a result of our neglection of fine structure terms in the Hamiltonian. The leading contribution to the ground state energy of the system beyond the Coulomb interactions already accounted for is the correction for the finite mass of the nucleus (assumed infinite in Eq. 1), which is proportional to $Z^2\mu/M$,[4] where Z is the nuclear charge, μ is the reduced mass fo the nucleus-electron system, and M is the nuclear mass. This correction gives a positive energy contribution, which may explain the convergence of our theoretical value below the experimentally-determined value.

In the case where we do not require m to be odd and use 64 states we find that, at the calculated minimal energy, $\kappa = 1.03$. Examining the form of the trial wavefunction, we see that κ acts as a screening constant for the electrons, reducing the effective nuclear charge felt. For a nuclear charge of $Z = 2$, this κ value corresponds to an effective charge experienced by each electron Z/κ of about 1.94. This is physically feasible; we would expect that each electron causes the other to experience a reduced effective nuclear charge, but given that they would be expected to be found further from one another than to the nucleus, we would not expect a significant deviation of the effective nuclear charge from $Z = 2$.

III. Extending to Three Electrons: The Slater Determinant

We now consider a system where the Hamiltonian is the same as Eq. 1, but we now have three electrons, so that $n = 3$. Before attempting to apply the Hylleraas method to systems with more than two electrons, we first examine the simpler method of applying the variational principle to a Slater determinant.

We write our trial wavefunction as the Slater determinant:

$$\psi = \frac{1}{\sqrt{6}} \begin{vmatrix} A_1 & A_2 & A_3 \\ \bar{A}_1 & \bar{A}_2 & \bar{A}_3 \\ B_1 & B_2 & B_3 \end{vmatrix}, \tag{9}$$

where each A_i corresponds to a 1s hydrogen-like wavefunction for the ith electron, $\bar{A}_i$ to the same function but with a negative spin factor, and B_i to the 2s hydrogen-like wavefunction with a positive spin factor. Writing the wavefunction as such a determinant allows us to satisfy the aforementioned spatial parity requirements.

In his work, Wilson uses the following wavefunctions:[6]

$$a = \sqrt{\frac{(Z-\sigma_1)^3}{\pi}} e^{-(Z-\sigma_1)r}, \tag{10}$$

$$b = k\left[\alpha r e^{-\frac{1}{2}(Z-\sigma_2)} - e^{-\frac{Z}{2}r} + \beta e^{-(Z-\sigma_1)r}\right], \tag{11}$$

where the a and b are A_i and B_i in Eq. 9 respectively, but without the spin factors. σ_1 and σ_2 are the variable screening constants for the K- and L-shell electrons respectively, and k and β are constants. α is initially fixed at $0.5(Z - \sigma_1)$.

We vary σ_1 and σ_2 using the downhill simplex method[7] to find the minimal energy, as illustrated in Fig. 2.

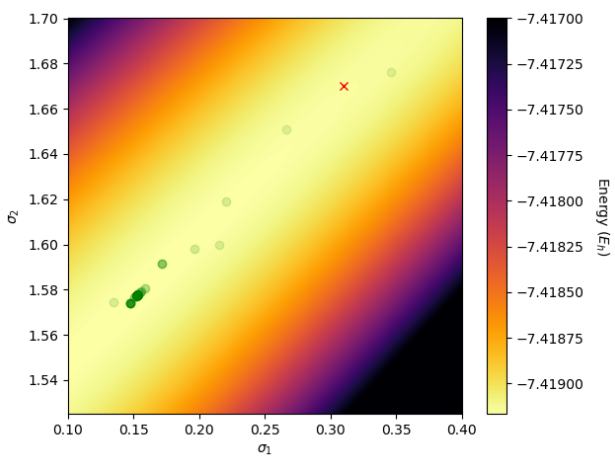

FIG. 2: The dependence of the calculated ground state energy on the electron screening constants. The constants obtained by Wilson are denoted by the red cross and the convergence of our determined value is visualised by increasing opacity of the green points.

Our determined value agrees with that obtained by Wilson of -7.419 E_h, but our screening constants of $\sigma_1 = 0.153$ and $\sigma_2 = 1.578$ do not agree with Wilson's determined values

of $\sigma_1 = 0.31$ and $\sigma_2 = 1.67$. The reason for this is most likely that the point of minimal energy is found within a shallow valley in which convergence is very slow, and given the lack of technology available to Wilson, his "tedious process" of finding the minimum would have succeeded in falling into this valley, but failed to converge towards the absolute minimum as in Fig. 2.

That such a valley of slow convergence is formed is of interesting physical significance, as it implies that increasing both screening constants results in only a very small change in the ground state energy. A possible explanation of this is that, although decreasing the strength of interaction of each electron with the nucleus would increase the potential energy of the system, the electrons would physically be further on average from the nucleus, and hence further apart from each other, which reduces the potential energy of each electron as a result of the presence of the other. The combination of these two effects then approximately cancel in this "valley".

Relaxing the constraint that α be fixed and applying the downhill simplex method alongside σ_1 and σ_2 in a three-dimensional parameter space, we again find a ground state energy of -7.419 E_h, suggesting that Wilson's assumption that α could remain fixed was justified.

This method fails to come reasonably close to the recommended value of -7.478 E_h.[8] In order to explain this discrepancy, we can consider various physical effects that the functions a and b do not account for.

We first consider the spatial overlap between the 1s and 2s orbitals. If the 2s electron is allowed to penetrate into the region occupied by the 1s electrons, it will experience a higher amount of the nuclear charge than outside this region.[9] However, the fact that the effective nuclear charge is increased also means that the electron would tend to be more accelerated away from the nucleus, meaning that it would not spend much time in the space occupied by the 2s orbital. It is therefore unlikely that this is the main reason for this discrepancy between the calculated and experimental value.

Another effect that we could consider is the polarisation of the 1s orbital (assumed in Wilson's work to be spherically-symmetric) due to the electron in the 2s orbital. While an expression for the change in the energy of the system as a result of this effect can be obtained through geometrical considerations,[10] it diverges for the $l = 0$ state. Instead, we could try to account for this effect through the use of computational methods,[11] but that is beyond the scope of this investigation.

IV. Applying the Hylleraas Method to Lithium

A. Constructing the Wavefunction

In order to obtain a more accurate value for the calculated ground state energy of lithium, James and Coolidge[12] propose a 10-term Hylleraas-style wavefunction that is a linear combination of terms of the form:

$$f(k,m,n,p,*) = (\rho_{23})^p[\zeta(k,m,n;\gamma,\delta) - \xi(k,m,n;\gamma,\delta)], \quad (12)$$

$$g(k,m,n,p,*) = (\rho_{12})^p\zeta(k,m,n;\gamma,\delta) - \rho_{13}\xi(k,m,n;\gamma,\delta), \quad (13)$$

$$h(k,m,n,p,*) = (\rho_{13})^p\zeta(k,m,n;\gamma,\delta) - \rho_{12}\xi(k,m,n;\gamma,\delta), \quad (14)$$

where δ is a constant fixed at 3 throughout this work and $*$ is a boolean argument where $\gamma = \gamma^* = 1.5$ if it is true and $\gamma = 0.65$ if it is false. ζ and ξ are:

$$\zeta(k,m,n;\gamma,\delta) = \sqrt{\frac{\kappa^9}{4\pi}}e^{-\gamma\rho_3+\delta\rho_2+\delta\rho_1}\rho_3^k\rho_2^m\rho_1^n, \quad (15)$$

$$\xi(k,m,n;\gamma,\delta) = \sqrt{\frac{\kappa^9}{4\pi}}e^{-\gamma\rho_2+\delta\rho_3+\delta\rho_1}\rho_2^k\rho_3^m\rho_1^n, \quad (16)$$

where $\rho_i = \kappa r_i$; $\rho_{ij} = \kappa r_{ij}$. We can therefore regard κ as a scaling constant.

The evaluation of the elements of the Hamiltonian and unity matrices requires the calculation of integrals of the form

$$I(j_1,j_2,j_3,j_{12},j_{23},j_{31};\alpha,\beta,\gamma) = \int d\vec{r}_1 d\vec{r}_2 d\vec{r}_3 r_1^{j_1} r_2^{j_2} r_3^{j_3} r_{12}^{j_{12}} r_{23}^{j_{23}} r_{31}^{j_{31}} e^{-\alpha r_1-\beta r_2-\gamma r_3}. \quad (17)$$

The integrands of the $\boldsymbol{N}$ elements given by Eq. 7 are trivially obtained by multiplying the corresponding terms of the wavefunction together (noting that each term is real), while the highly complicated integrands of the $\boldsymbol{H}$ elements given by Eq. 6 may be obtained by symbolic computation[13] using the explicit differential forms of $\hat{T}$ and $\hat{V}$. The evaluation of such integrals is highly non-trivial and warrants a discussion in its own right.

B. Evaluation of the Three-Electron Integral

While Wilson discusses the evaluation of integrals of the form of Eq. 17, such integrals have received more attention since, and a particularly elegant method is presented by Drake and Yan, the asymptotic expansion method.[14] We have chosen this method over that presented by Wilson due to its higher simplicity with any set of arguments of I, as oppose to Wilson's method which only discusses certain special cases. This involves the evaluation of the three-electron integral I using the sum

$$I = \sum_{q=0}^{\infty} T(q), \quad (18)$$

where $T(q)$ is a weighted sum of integrals of the form

$$W(l,m,n;\alpha,\beta,\gamma) = \int_0^\infty dx x^l e^{-\alpha x} \int_0^\infty dy y^m e^{-\beta y} \int_0^\infty dz z^n e^{-\gamma z}, \quad (19)$$

which Drake and Yan show can be evaluated using the expression

$$W(l,m,n;\alpha,\beta,\gamma) = \frac{l!}{(\alpha+\beta+\gamma)^{l+m+n+3}} \times \sum_{p=0}^{\infty} \frac{(l+m+n+p+2)!}{(l+1+p)!\,(l+m+p+2)} \left(\frac{\alpha}{\alpha+\beta+\gamma}\right)^p \times {}_2F_1\left(1, l+m+n+p+3; l+m+p+3; \frac{\alpha+\beta}{\alpha+\beta+\gamma}\right), \quad (20)$$

where ${}_2F_1$ is the hypergeometric function.

In order to evaluate I, the asymptotic-expansion method is applied where the infinite sum is split into two parts:

$$I = \sum_{q=0}^{N} T(q) + \sum_{N+1}^{\infty} T(q) \equiv I_{0,N} + I_{N+1,\infty}. \quad (21)$$

We assume that $I_{N+1,\infty}$ can be approximated by

$$I_{N+1,\infty} \approx \sum_{i=0}^{\Lambda} A_i \zeta_N(i+\lambda) \equiv I_{N+1,\Lambda}, \quad (22)$$

where $\zeta_N = \sum_{j=N+1}^{\infty} 1/j^i$, $\lambda = (j_{12}+1)/2+(j_{23}+1)/2+(j_{31}+1)/2+4$, and each A_i is determined as described in the Appendix. In this work, we use Drake and Yan's recommended values of $N = 20$ and $\Lambda = 7$.

A major question this method raises is that of how many terms should be used in Eq. 20. This is of high importance as the evaluation of this expression takes up most of the computing time in evaluating the Hamiltonian and unity matrix elements, so we would ideally evaluate the lowest number of terms that does not affect the accuracy of the final result.

Fig. 3 shows the evaluated asymptotic expansion as a function of the minimum difference between the successive terms of the sum in Eq. 20 in order for the evaluation to terminate, t (the convergence tolerance). We see that the asymptotic expansion part of I appears to converge to a constant value, before oscillating orders of magnitude above its steady value, before again settling on a steady value. In this section, we use Drake and Yan's test case of $I = I(0,0,0,-1,-1,-1;1,1,1)$ with $N = 20$ and $\Lambda = 7$.

Also shown in Fig. 3 is the calculated value of $T(q)$ as a function of the convergence tolerance for $q = N - \Lambda, N - \Lambda + 1, \ldots, N = 13, 14, \ldots, 20$, with decreasing opacity corresponding to increasing q. We see that each $T(q)$ is constant in the region of low convergence tolerance

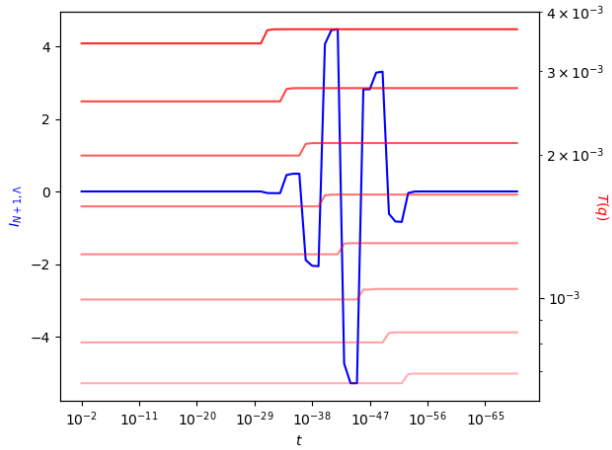

FIG. 3: The evaluated value of the asymptotic expansion term (shown by a blue line) and the change in $T(q)$ (shown by a red line, with an decreasing opacity with increasing q) as a function of the convergence tolerance t.

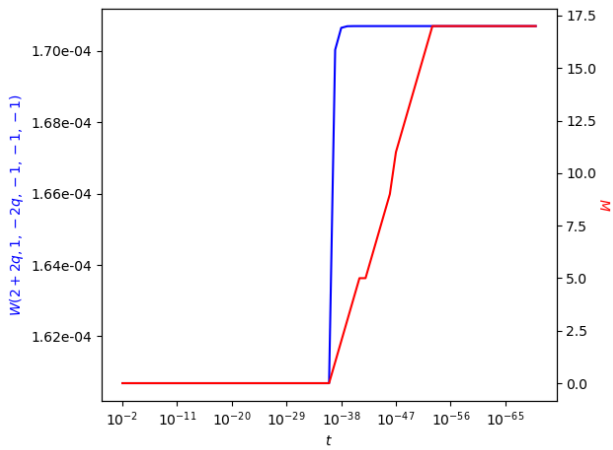

FIG. 4: The evaluated W (shown by a blue line) for $q = 15$ for the Drake and Yan test case and the corresponding minimum value of M required as a function of the convergence tolerance.

before the oscillations, before they successively "jump" as q increases, before being relatively constant again. It is then easily seen that the oscillations start where $T(q)$ sees a sudden jump for $q = 13$ and ends at the jump where $q = 20$.

To examine the cause of these sudden rises, we note that, in this test case, $T(q) \propto W(2+2q, 1, -2q; -1, -1, -1)$. Examining Fig. 4, we see that each of these rises corresponds to the value of the minimum upper limit of p in Eq. 20, M, required to achieve convergence within the convergence tolerance increasing above 0, meaning that more than just the first term in the sum is being evaluated. The sum then quickly converges.

Looking at Fig. 5, we note that the A_i coefficients blow up in the region in which the asymptotic expansion is unstable. This is due to the fact that, at higher convergence tolerances before this unstable region, the values of $T(q)$ remain constant as only the first term in the corresponding W's are evaluated. As we enter the region at which the asymptotic expansion is unstable with respect to small changes in the convergence criterion, the evaluation of $T(q)$ requires evaluations of Eq. 20 where M rises above

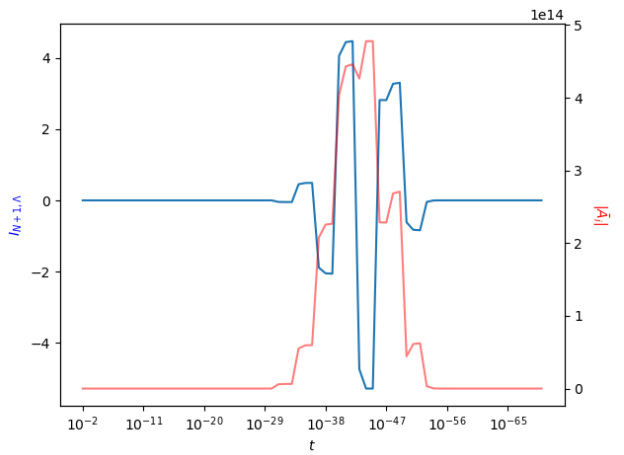

FIG. 5: Asymptotic expansion (represented by the blue line), and the absolute value of the average of coefficient A_i in Eq. 22 over $i = 0$ to Λ as a function of the convergence criterion t.

zero, causing sharp jumps in their values, while others still have $M = 0$. This mixed state results in A_i values that blow up, resulting in the asymptotic expansion oscillating several orders of magnitude about its value in the limit of a small convergence criterion. The oscillations stop as the convergence criterion decreases even further until M rises above 0 for all values of q.

The remedy to this problem is to take the limit that the convergence criterion tends to zero, as extensive testing has found that M remains constant in this limit. Although in reality the difference between two terms of the sum in Eq. 20 will only become zero in the limit of $M \to \infty$, in practice the difference becomes so small that a computer cannot distinguish it from zero, so M remains finite, as can be seen in the plateau of M as t decreases in Fig. 4. We have therefore found a way to vary M on a case-by-case basis such that it is not too computationally-expensive for quickly-converging sums, but remains accurate for sums that are slower to converge and hence a higher value of M.

C. Evaluation of the Lithium Ground State Energy

Once we have built up the Hamiltonian and unity matrices, we can solve the eigenvalue equation to obtain the ground state energy of lithium. Ideally, we would like to vary several parameters κ, δ, γ, and γ^*. However, as computing the Hamiltonian and unity matrices for just one set of parameters takes about 30 minutes, varying the parameters pseudo-continuously as done in Fig. 2 and carrying out multiple-variable minimisation would not be feasible with the available computing power. Instead, as suggested in the corresponding literature, we set $\delta = 3$, $\gamma = 0.65$, and $\gamma^* = 1.5$, and only minimise the energy with respect to κ.

Carrying out this procedure with the 10-term wavefunction, we obtain a ground state energy of lithium of -7.463 E_h, with $\kappa = 0.95$, as can be seen in Fig. 6, which is still not very close to the recommended value of -7.478 E_h, but is closer than our value obtained using Wilson's Slater determinant method. It is also not close to James and Coolidge's calculated value of -7.476 E_h, corresponding to $\kappa = 0.93$. The reason for the discrepancy between our calculated ground state energy using the 10-term wavefunction and that obtained by James and Coolidge is unclear, but could have several explanations.

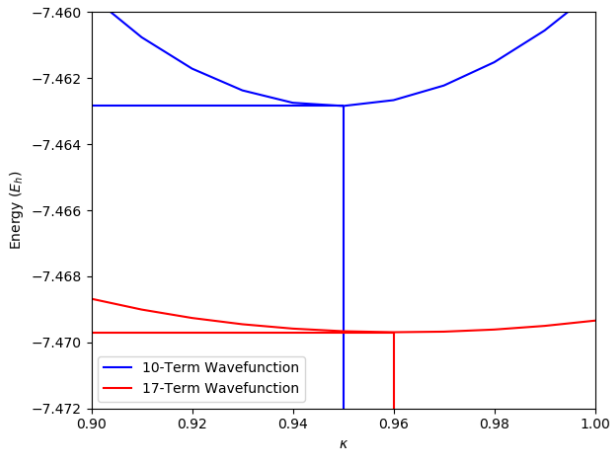

FIG. 6: The variation in the ground state energy of lithium as a function of the variable parameter κ for the 10-term wavefunction (shown by the red line) and the 17-term wavefunction (shown by the blue line).

One explanation is that we use a different method to evaluate the three-electron integral than that in the paper, which has only been tested for the relatively simple case of $j_1 = j_2 = j_3 = 0$; $j_{12} = j_{23} = j_{31} = -1$; $\alpha = \beta = \gamma = 1$. More testing before applying it to the more complicated integrals when using our trial wavefunction would be beneficial. Additionally, we cut off the sums at $N = 20$ and $\Lambda = 7$. If we used higher values, we would obtain more accurate values of the three-electron integrals, and hence of the eigenenergies.

Another possible explanation for the discrepancy is that the way in which we calculate the ground state energy from the eigenvalue equation $\hat{H}\psi = E\psi$, where ψ is the Hylleraas-style wavefunction and E is the system's (ground state) energy, differs. James and Coolidge provide a coefficient for each term in the trial wavefunction, but our method of obtaining energy eigenvalues using Eq. 8 does not require the explicit knowledge of these coefficients as they are implicitly determined in our method of extracting the ground state eigenenergy so that they vary with the choice of κ. Why this leads to a discrepancy, however, is unclear. We would expect that, if the weights of each term are changing, the value of κ which minimises the ground state energy should be different from that obtained by James and Coolidge (which we observe) in order to make the linear combination of functions physically equivalent. The physical interpretation of this is that, while varying κ stretches the units of length, varying the coefficients changes the degree to which each term in the trial wavefunction contributes towards the overall wavefunction, so we should expect that these two effects cancel in order to yield the same ground state energy.

In an attempt to obtain a ground state energy closer to the experimental value, we also try using the 17-term wavefunction described by James and Coolidge, a linear combination of the same f, g, and h as used in the 10-term trial wavefunction, but using a choice of terms that is more

physically-motivated, attempting to account for effects such as inter-shell polarisation. Referring to Fig. 6, we can see that this provides a ground-state energy closer to the experimental value than the 10-term function for all values of κ, finding a value of -7.470 E_h, with $\kappa = 0.96$. While this is closer to the value found in the literature, it still does not come satisfactorily close to it, again possibly for the reasons discussed above. It should also be noted that the value obtained for κ using this wavefunction is further from that found in the literature than that using the 10-term wavefunction, but, as previously discussed, with our different method of obtaining the eigenenergies, we should not expect the values of κ to agree with those obtained by James and Coolidge. Additionally, as more terms are added, a small change in κ will have less effect on the evaluated energy (i.e. the curves in Fig. 6 get flatter), so we should expect that, as we increase the number of terms, κ should be determined to a lower accuracy.

V. Conclusions and Further Investigations

We have found that applying the variational method to calculate the ground state energy of helium using Hyleraas' basis states has given a good agreement with experimental values. We have not been so successful extending this to the lithium atom, though this method has proved more successful than the Slater determinant method. The results may be improved through the investigation of different methods to extract the system's eigenenergies, keeping the coefficients of the terms in the Hylleraas wavefunction constant as parameters were varied, as per James and Coolidge's work.

The main challenge of the use of Hyleraas-style wavefunctions and the variational method to obtain the ground state energy of many-electron systems is the evaluation of the three-electron integrals in calculating the Hamiltonian and unity matrix elements. We have tested Drake and Yan's asymptotic expansion method and found that, within the precision limitations of modern computers, setting the convergence criterion for additional terms in the sum obtained by Drake and Yan to evaluate the W integral to zero is satisfactory. However, future investigations could attempt to use more terms in the asymptotic expansion method (i.e. higher values of N and Λ) in order to obtain more accurate values of such integrals.

In minimising the energy obtained through James and Coolidge's trial wavefunctions, we only varied κ due to the large amount of time required to evaluate the three-electron integrals. Ideally, we would also vary δ, γ, and γ^* and use minimisation in parameter space in a similar manner to the variation of the screening constants shown in Fig. 2. This would, however, only be feasible if we used a much more powerful computer, or a method of evaluating the three-electron integrals much more quickly was developed.

Finally, another natural next step to this investigation would be to modify the Hamiltonian to account for the finite nuclear mass, relativistic effects, and quantum electrodynamical effects, which account for much of the discrepancy between literature and experimental values. It may also be worth investigating different, possibly new methods of obtaining theoretical values for the ground state energies of many-electron systems beyond the early 20th-century methods of the Slater determinant and the variational method with Hylleraas-style wavefunctions.

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Appendix: Determination of A_i Coefficients

We assume that, for some integer Λ, each $T(q)$ is given exactly by the expression

$$T(q) = \sum_{i=0}^{\Lambda} \frac{A_i}{q^{i+\lambda}},$$

then we calculate $T(q)$ for $q = N - \Lambda$, $q = N - \Lambda + 1, \ldots,$ $q = N$, which we can turn into the matrix equation

$$\begin{bmatrix} T(N-\Lambda) \\ T(N-\Lambda+1) \\ \vdots \\ T(N) \end{bmatrix} = \begin{bmatrix} \frac{1}{(N-\Lambda)^{\lambda}} & \frac{1}{(N-\Lambda)^{\lambda+1}} & \cdots & \frac{1}{(N-\Lambda)^{\lambda+\Lambda}} \\ \frac{1}{(N-\Lambda+1)^{\lambda}} & \frac{1}{(N-\Lambda+1)^{\lambda+1}} & \cdots & \frac{1}{(N-\Lambda+1)^{\lambda+\Lambda}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{N^{\lambda}} & \frac{1}{N^{\lambda+1}} & \cdots & \frac{1}{N^{\lambda+\Lambda}} \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ \vdots \\ A_{\Lambda} \end{bmatrix},$$

or

$$\boldsymbol{T} = \boldsymbol{M}\boldsymbol{a},$$

which can easily be solved for $\boldsymbol{a}$ pre-multiplying each side by $\boldsymbol{M}^{-1}$.